

NLU course project - Part 2A and Part 2B

Emanuele Ricca (258437)

University of Trento

emanuele.ricca@studenti.unitn.it

1. Introduction

This report investigates joint multi-task learning for intent classification and slot filling on the Air Travel Information System (ATIS) dataset. The objective is to evaluate how architectural modifications and transfer learning affect performance on both tasks within a shared model.

The first part focuses on improving the baseline architecture provided for the course through incremental modifications. Model capacity is progressively increased by adjusting hidden dimensions, attention heads, layer depth, and feed-forward dimensions. In addition, dropout regularization is introduced and evaluated to determine its effect on generalization.

The second part compares these models against fine-tuned pre-trained transformers. Both GPT2 and BERT-base-uncased are adapted to the joint intent classification and slot filling setting. Particular attention is given to handling sub-tokenization during slot prediction and to the different strategies required for intent classification in encoder-only and decoder-only architectures. Generative AI tools were used only for debugging support, code verification, and document formatting.

2. Implementation Details

The multi-task architecture uses a shared contextual representation matrix $H \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ that is processed by two task-specific prediction heads. For slot filling, each token representation H_i is projected onto the slot label space. For intent classification, a sentence-level representation is extracted from the backbone model and passed to a dedicated classifier.

The representation used for intent classification depends on the underlying architecture. For the custom model and BERT, the hidden state associated with the [CLS] token ($\mathbf{h}_{[\text{CLS}]} = H[:, 0]$) is used as the sentence representation [1]. For GPT2, which does not contain a classification token and operates autoregressively, the hidden state of the final non-padding token is selected using the attention mask ($\mathbf{h}_{\text{final}} = H[B, \sum(\text{mask}) - 1]$).

Training is performed using a joint objective consisting of the sum of the intent classification loss and slot filling loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{intent}} + \mathcal{L}_{\text{slot}} \quad (1)$$

To address the sub-tokenization introduced by WordPiece and Byte-Pair Encoding tokenizers, slot labels are assigned only to the first sub-token of each word. All remaining sub-tokens and padding positions are assigned the ignore index (SLOT_IGNORE_INDEX = -100), preventing them from contributing to the loss or gradient updates. All models are optimized using AdamW [2] with linear learning-rate warmup and early stopping.

3. Results and Evaluation

Models are evaluated on the ATIS dataset using Intent Classification Accuracy and CoNLL-style Slot F1. Table 1 reports the results obtained by incrementally modifying the baseline architecture, while Table 2 summarizes the performance of the fine-tuned pre-trained models.

Table 1: Incremental optimization results for Part 2.A (From Scratch).

Incremental Modification Variant	Intent Acc.	Slot F1
Provided Baseline ($d_{\text{model}} = 20, L = 1, H = 1, \text{FF}_{\text{dim}} = 20$)	0.8847	0.8093
+ Capacity Scale Up ($d_{\text{model}} = 64, \text{FF}_{\text{dim}} = 128$)	0.9250	0.8834
+ Full Capacity Scaling ($d_{\text{model}} = 128, \text{FF}_{\text{dim}} = 192$)	0.9317	0.8839
+ Multi-Head Expansion ($H = 2, d_{\text{model}} = 128$)	0.9183	0.8721
+ Deep Layer Stacking Expansion ($L = 4, d_{\text{model}} = 64$)	0.9037	0.8674
+ Depth Scaling Layer Setup ($L = 2, d_{\text{model}} = 64$)	0.9272	0.8714
+ Deep Multi-Head Scale Up ($L = 4, H = 2, d_{\text{model}} = 64$)	0.9306	0.8884
+ Deep Regularized Layout (Dropout _{reg} = 0.1, $L = 4, H = 2, d_{\text{model}} = 128$)	0.9395	0.8906

The baseline model achieves an Intent Accuracy of 0.8847 and a Slot F1 score of 0.8093. Increasing the model dimensions provides the largest improvement, with both tasks benefiting from the additional representational capacity. Scaling d_{model} from 20 to 64 and later to 128 consistently improves performance, suggesting that model width is one of the main factors affecting results on ATIS.

Increasing the number of attention heads or the number of layers in isolation does not always produce further gains. Some deeper or more complex configurations show slightly lower scores, likely because of the relatively small size of the dataset. A more balanced setup combining increased capacity, additional layers, and multiple attention heads performs better than these isolated modifications.

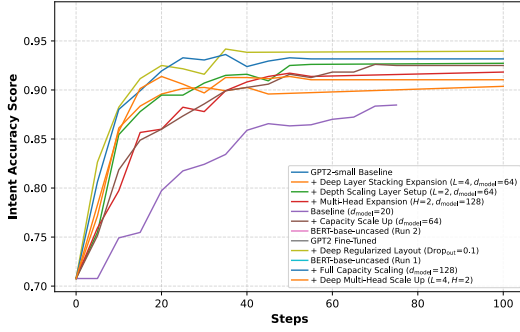
The best model trained from scratch combines larger dimensions, deeper layers, multiple attention heads, and an output dropout of 0.1. This configuration reaches an Intent Accuracy of 0.9395 and a Slot F1 score of 0.8906, representing the strongest performance among the custom architectures.

Table 2: Multi-task transfer fine-tuning results for Part 2.B.

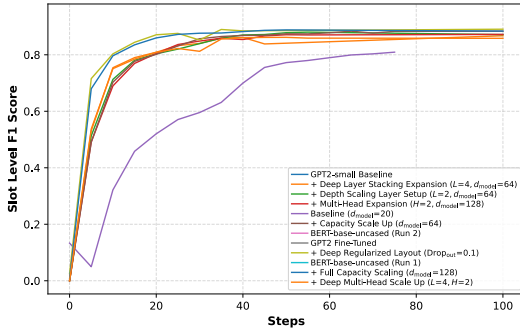
Pre-trained Backbone Model	Intent Acc.	Slot F1
GPT2-small Initial Trial Configuration	0.9474	0.8822
GPT2 Fully Fine-Tuned Setup	0.9586	0.8949
BERT-base-uncased Architecture (Run 1)	0.9653	0.9307
BERT-base-uncased Architecture (Run 2)	0.9765	0.9332

Fine-tuning pre-trained transformer models produces substantially better results than any configuration trained from scratch. Even the initial GPT2-small setup exceeds the performance of the best custom architectures after only a few epochs of training. Fully fine-tuning GPT2 further improves performance, reaching an Intent Accuracy of 0.9586 and a Slot F1 score of 0.8949.

Among all evaluated models, BERT-base-uncased achieves the highest overall results. The best run obtains an Intent Accuracy of 0.9765 and a Slot F1 score of 0.9332. These improvements are likely related to BERT’s bidirectional attention mechanism, which allows token representations to incorporate information from both left and right context. This is particularly useful for slot filling, where predictions often depend on surrounding words. In contrast, GPT2 relies on a causal attention mask and therefore only has access to preceding context when building token representations.



(a) Multi-Task Intent Accuracy Evaluation Profile (Combined Test Set)



(b) Multi-Task Slot F1 Sequence Tagging Evaluation Profile (Combined Test Set)

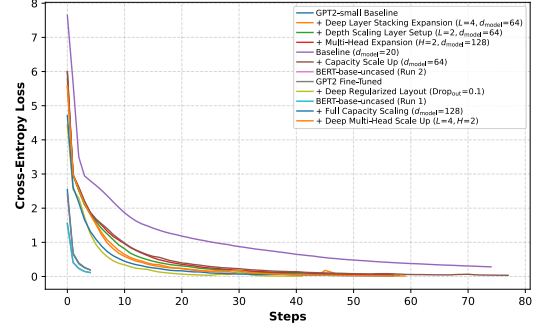
Figure 1: *Comparative performance metrics between custom models built from scratch and pre-trained backbone transformers, illustrating intent classification trajectories (a) and sequence-tagging token F1 accuracy gains (b).*

Figure 1 summarizes the performance differences between the models trained from scratch and the fine-tuned transformer backbones. The pre-trained models consistently achieve higher scores on both tasks, highlighting the benefit of transfer learning and large-scale pre-training.

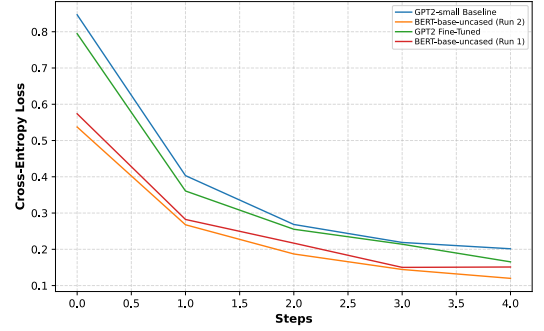
The advantage is particularly visible for slot filling in Figure 1b, where BERT-base-uncased establishes a clear performance gap over the custom architectures. While the scratch-trained models improve through architectural scaling, they do not reach the same level of performance as the pre-trained alternatives.

Figure 2 compares the training and validation loss curves of all evaluated systems. Models trained from scratch require more optimization steps before reaching stable performance, whereas the fine-tuned transformer models converge much more rapidly.

This behaviour reflects the advantage of transfer learning: the pre-trained models start from useful linguistic representations and therefore require fewer updates to adapt to the ATIS



(a) Cross-Entropy Convergence Profile (Combined Training Track)



(b) Cross-Entropy Generalization Profile (Combined Validation Track)

Figure 2: *Joint optimization loss trajectories across all evaluated configurations on both training (a) and validation (b) splits.*

tasks. As a result, both GPT2 and BERT achieve lower validation losses and more stable convergence than the custom architectures.

4. References

- [1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *NAACL*, 2019.
- [2] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *ICLR*, 2018.